

Lab04_Caparrós_ME315A

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Hands-On

Exercício 1:

Carregue os pacotes readxl e tidyverse.

```
library(readxl)
library(tidyverse)
```

Exercício 2:

Crie uma variável chamada fname que contenha o caminho para o arquivo WDIEXCEL.xlsx, disponível no seu diretório de dados. Confirme que o R identifica a existência do arquivo utilizando o comando file.exists

```
fname <- "WDIEXCEL.xlsx"

file.exists(fname)
```

```
[1] TRUE
```

Exercício 3:

Leia a planilha “Country” e armazene no objeto countries.

```
countries <- read_excel(fname, sheet = "Country")

countries
```

```
# A tibble: 263 x 30
  `Country Code` `Short Name`      `Table Name`  `Long Name`  `2-alpha code`
  <chr>          <chr>          <chr>         <chr>        <chr>
1 ABW           Aruba           Aruba         Aruba        AW
2 AFG           Afghanistan  Afghanistan  Islamic St~  AF
3 AGO           Angola         Angola        People's R~  AO
4 ALB           Albania        Albania       Republic o~  AL
5 AND           Andorra        Andorra       Principali~  AD
6 ARB           Arab World     Arab World     Arab World    1A
7 ARE           United Arab Emirates United Arab E~ United Ara~  AE
8 ARG           Argentina      Argentina     Argentine ~  AR
9 ARM           Armenia        Armenia       Republic o~  AM
10 ASM          American Samoa  American Samoa American S~  AS
# i 253 more rows
# i 25 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <chr>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, ...
```

Exercício 4:

Mantenha apenas os registros de 5 países (Brasil e outros 4 de sua escolha).

```
countries <- countries |>
  filter(`Short Name` %in% c(
    "Brazil",
    "Argentina",
    "Chile",
    "Mexico",
    "United States"
  ))
countries
```

```
# A tibble: 5 x 30
  `Country Code` `Short Name`      `Table Name`  `Long Name`  `2-alpha code`
  <chr>          <chr>          <chr>         <chr>        <chr>
1 ARG           Argentina      Argentina     Argentine Republic AR
2 BRA           Brazil        Brazil        Federative Republic~ BR
3 CHL           Chile         Chile         Republic of Chile  CL
```

```

4 MEX                Mexico          Mexico          United Mexican Stat~ MX
5 USA                United States United States United States of Am~ US
# i 25 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <chr>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, `Alternative conversion factor` <chr>,
#   `PPP survey year` <chr>, `Balance of Payments Manual in use` <chr>, ...

```

Exercício 5:

Leia a planilha “Data” e armazene no objeto WDI.

```
WDI <- read_excel(fname, sheet = "Data")
```

```
WDI
```

```

# A tibble: 422,136 x 63
  `Country Name` `Country Code` `Indicator Name` `Indicator Code` `1960` `1961`
  <chr>          <chr>          <chr>          <chr>          <dbl> <dbl>
1 Arab World    ARB                2005 PPP conver~ PA.NUS.PPP.05      NA      NA
2 Arab World    ARB                2005 PPP conver~ PA.NUS.PRVT.PP.~    NA      NA
3 Arab World    ARB                Access to clean~ EG.CFT.ACCS.ZS     NA      NA
4 Arab World    ARB                Access to elect~ EG.ELC.ACCS.ZS     NA      NA
5 Arab World    ARB                Access to elect~ EG.ELC.ACCS.RU.~    NA      NA
6 Arab World    ARB                Access to elect~ EG.ELC.ACCS.UR.~    NA      NA
7 Arab World    ARB                Account ownersh~ FX.OWN.TOTL.ZS     NA      NA
8 Arab World    ARB                Account ownersh~ FX.OWN.TOTL.FE.~    NA      NA
9 Arab World    ARB                Account ownersh~ FX.OWN.TOTL.MA.~    NA      NA
10 Arab World   ARB                Account ownersh~ FX.OWN.TOTL.OL.~    NA      NA
# i 422,126 more rows
# i 57 more variables: `1962` <dbl>, `1963` <dbl>, `1964` <dbl>, `1965` <dbl>,
#   `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>, `1970` <dbl>,
#   `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>, `1975` <dbl>,
#   `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>, `1980` <dbl>,
#   `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>, `1985` <dbl>,
#   `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>, `1990` <dbl>, ...

```

Exercício 6:

Selecione apenas as colunas:

- Country Code
- Indicator Code
- E todos os anos de 1960-2018

```
WDI <- WDI |>
  select(`Country Code`, `Indicator Code`, `1960`:`2018`)

WDI
```

```
# A tibble: 422,136 x 61
  `Country Code` `Indicator Code` `1960` `1961` `1962` `1963` `1964` `1965`
  <chr>          <chr>          <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 ARB           PA.NUS.PPP.05      NA     NA     NA     NA     NA     NA
2 ARB           PA.NUS.PRVT.PP.05  NA     NA     NA     NA     NA     NA
3 ARB           EG.CFT.ACCS.ZS     NA     NA     NA     NA     NA     NA
4 ARB           EG.ELC.ACCS.ZS     NA     NA     NA     NA     NA     NA
5 ARB           EG.ELC.ACCS.RU.ZS   NA     NA     NA     NA     NA     NA
6 ARB           EG.ELC.ACCS.UR.ZS   NA     NA     NA     NA     NA     NA
7 ARB           FX.OWN.TOTL.ZS     NA     NA     NA     NA     NA     NA
8 ARB           FX.OWN.TOTL.FE.ZS   NA     NA     NA     NA     NA     NA
9 ARB           FX.OWN.TOTL.MA.ZS   NA     NA     NA     NA     NA     NA
10 ARB          FX.OWN.TOTL.OL.ZS   NA     NA     NA     NA     NA     NA
# i 422,126 more rows
# i 53 more variables: `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>,
#   `1970` <dbl>, `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>,
#   `1975` <dbl>, `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>,
#   `1980` <dbl>, `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>,
#   `1985` <dbl>, `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>,
#   `1990` <dbl>, `1991` <dbl>, `1992` <dbl>, `1993` <dbl>, `1994` <dbl>, ...
```

Exercício 7:

Confirme se os dados wm WDI estão em formato tidy. Se não estiverem, transforme-os para tal formato. Faça de forma que exista uma coluna referente ao ano chamada YEAR.

```
WDI <- WDI |>
  pivot_longer(
    cols = `1960`:`2018`,
    names_to = "YEAR",
    values_to = "VALUE"
  )
```

```
WDI
```

```
# A tibble: 24,906,024 x 4
  `Country Code` `Indicator Code` YEAR  VALUE
  <chr>          <chr>          <chr> <dbl>
1 ARB           PA.NUS.PPP.05      1960    NA
2 ARB           PA.NUS.PPP.05      1961    NA
3 ARB           PA.NUS.PPP.05      1962    NA
4 ARB           PA.NUS.PPP.05      1963    NA
5 ARB           PA.NUS.PPP.05      1964    NA
6 ARB           PA.NUS.PPP.05      1965    NA
7 ARB           PA.NUS.PPP.05      1966    NA
8 ARB           PA.NUS.PPP.05      1967    NA
9 ARB           PA.NUS.PPP.05      1968    NA
10 ARB          PA.NUS.PPP.05      1969    NA
# i 24,906,014 more rows
```

Exercício 8:

Confirme se a coluna referente ao ano está em formato inteiro.

```
WDI$YEAR <- as.integer(WDI$YEAR)

is.integer(WDI$YEAR)
```

```
[1] TRUE
```

Exercício 9:

Para o objeto WDI, mantenha apenas os registros dos países listados em countries. Selecione as colunas:

- Country Code
- Short Name
- YEAR
- SP.DYN.CBRT.IN

```

WDI <- WDI |>
  filter(
    `Country Code` %in% countries$`Country Code`,
    `Indicator Code` == "SP.DYN.CBRT.IN"
  ) |>
  left_join(
    countries |>
      select(`Country Code`, `Short Name`),
    by = "Country Code"
  ) |>
  select(
    `Country Code`,
    `Short Name`,
    YEAR,
    `SP.DYN.CBRT.IN` = VALUE
  )

```

WDI

```

# A tibble: 295 x 4
  `Country Code` `Short Name`  YEAR SP.DYN.CBRT.IN
  <chr>          <chr>      <int>      <dbl>
1 ARG           Argentina    1960        23.7
2 ARG           Argentina    1961        23.5
3 ARG           Argentina    1962        23.2
4 ARG           Argentina    1963        23.0
5 ARG           Argentina    1964        22.9
6 ARG           Argentina    1965        22.7
7 ARG           Argentina    1966        22.6
8 ARG           Argentina    1967        22.5
9 ARG           Argentina    1968        22.5
10 ARG          Argentina    1969        22.6
# i 285 more rows

```

Exercício 10:

Ordene o objeto de forma que, para cada país, as informações estejam ordenadas por ano.

```

WDI <- WDI |>
  arrange(`Short Name`, YEAR)

```

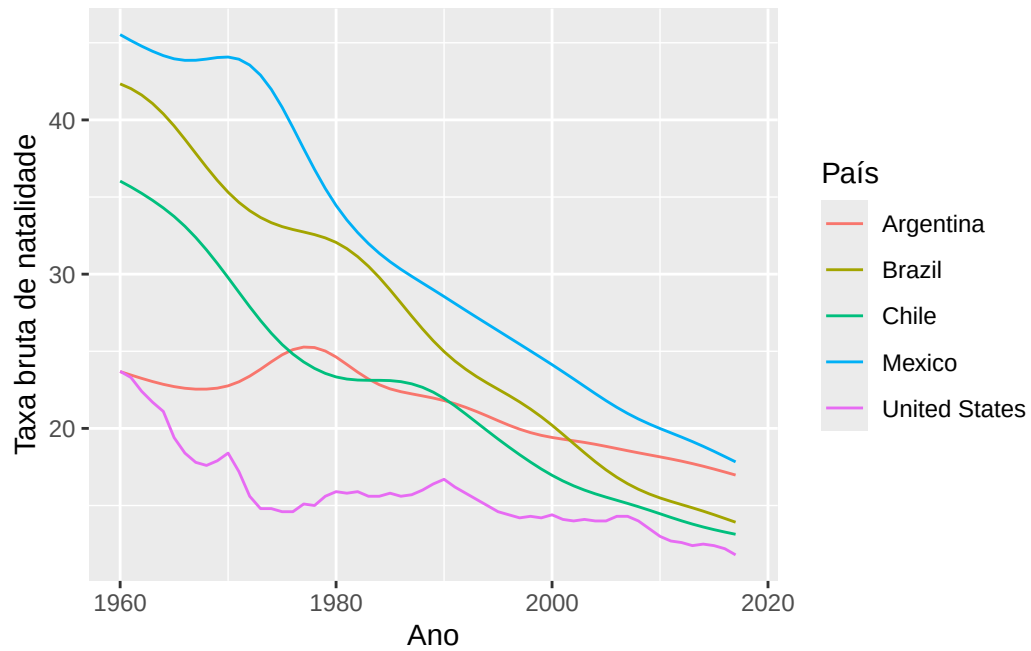
WDI

```
# A tibble: 295 x 4
  `Country Code` `Short Name` YEAR SP.DYN.CBRT.IN
  <chr>          <chr>      <int>      <dbl>
1 ARG           Argentina  1960       23.7
2 ARG           Argentina  1961       23.5
3 ARG           Argentina  1962       23.2
4 ARG           Argentina  1963       23.0
5 ARG           Argentina  1964       22.9
6 ARG           Argentina  1965       22.7
7 ARG           Argentina  1966       22.6
8 ARG           Argentina  1967       22.5
9 ARG           Argentina  1968       22.5
10 ARG          Argentina  1969       22.6
# i 285 more rows
```

Exercício 11:

Construa um gráfico de linhas, usando o pacote ggplot2, que tenha no eixo X a variável YEAR, no eixo Y a variável SP.DYN.CBRT.IN e que cada linha corresponda a um país diferente.

```
ggplot(WDI, aes(
  x = YEAR,
  y = `SP.DYN.CBRT.IN`,
  group = `Short Name`,
  color = `Short Name`
)) +
  geom_line() +
  labs(
    x = "Ano",
    y = "Taxa bruta de natalidade",
    color = "País"
  )
```



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-09 23:38:21 -03"
```